Electrical Network Analysis - Lab

Open Ended Lab Report

Passive High Pass Filter

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Introduction

A high-pass filter is an electronic filter that passes signals with a frequency higher than a certain cutoff frequency and attenuates signals with frequencies lower than the cutoff frequency. The amount of attenuation for each frequency depends on the filter design. A High Pass Filter is the exact opposite to the low pass filter circuit as the two components have been interchanged with the filters output signal now being taken from across the resistor. Whereas the low pass filter only allowed signals to pass below its cut-off frequency point, ƒc, the passive high pass filter circuit as its name implies, only passes signals above the selected cut-off point, ƒc eliminating any low frequency signals from the waveform. The Passive High Pass Filter is the exact opposite to the low pass filter. This filter has no output voltage from DC (0Hz), up to a specified cut-off frequency (ƒc) point.

For RC high pass filter, the cut-off frequency, ƒc, is calculated as

$$f\mathrm{c} = \frac{1}{2\pi RC}$$

Similarly, for RL high pass filter, the cut-off frequency, ƒc, is calculated as

$$f\mathrm{c} = \frac{R}{2\pi L}$$

Problem Analysis

We have to design a RL and RC high pass filter circuit for a cut-off frequency of 5kHz. We use a 150mH inductor for RL circuit and 10nF capacitor for RC circuit. We have to calculate the value of resistors for these circuits.

The specified cutoff frequency is 5kHz, which means that the filters should allow frequencies higher than 5kHz to pass through relatively un-attenuated, while attenuating lower frequencies.

Calculations

RL-Circuit:-

RC-Circuit:-

Design Requirements

Following equipment is needed for these circuits:-

- 10nF Capacitor
- 150mH Inductor
- 3.2kΩ Resistor
- 4.7kΩ Resistor
- Oscilloscope
- Function Generator
- Breadboard

Circuit Diagrams

RL-Circuit:-

RC-Circuit:-

Theoretical Results

RL-Circuit:-

f (Hz)	Vin (Vpp)	|Vout| V	θ (°)	Av

RC-Circuit:-

f (Hz)	Vin (Vpp)	|Vout| V	θ (°)	Av

Measured Results

RL-Circuit:-

f (Hz)	Vin (Vpp)	|Vout| V	θ (˚)	Av

RC-Circuit:-

f (Hz)	Vin (Vpp)	|Vout| V	θ (˚)	Av

Graphs

Conclusion

In conclusion, this experiment focused on the design and implementation of RL and RC passive high pass filters with a specified cutoff frequency of 5kHz. By utilizing a 150mH inductor for the RL circuit and a 10nF capacitor for the RC circuit, we calculated the resistor values necessary to achieve the desired cutoff frequency.

Additionally, we extended the experiment by calculating and analyzing the output voltage (Vout) at different frequencies, providing valuable insights into the filters' frequency response characteristics.

RL and RC high pass filters find applications in various electronic systems and signal processing scenarios due to their ability to allow higher frequencies to pass through while attenuating lower frequencies. Here are a few common applications for both RL and RC high pass filters:-

1. Audio Systems
2. Communication Systems
3. Crossover Networks in Speakers
4. Instrumentation and Measurement Systems
5. Signal Processing in Audio Equalization
6. Biomedical Devices
7. Radio Frequency (RF) Circuits
8. Antenna Systems
9. Power Supply Filtering
10. Seismic Signal Processing

These applications highlight the versatility of RL and RC high pass filters in various domains, contributing to improved signal quality, noise reduction, and optimized system performance.